

Improve performance of existing online trackers for small object detection

UAV videos

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- **Challenges** faced in detection:
 - Tiny objects → few pixels
 - Clutter, motion blur, scale variation
 - Unreliable detections
- Unreliable detections lead to:
 - Missed detections
 - Identity switches
 - Poor tracking performance

Research Gap

Existing methods are not optimized for small object detection in UAV scenarios, resulting in poor detection quality and inconsistent identity association even with tiling and ReID.

Spatial Detection Challenges

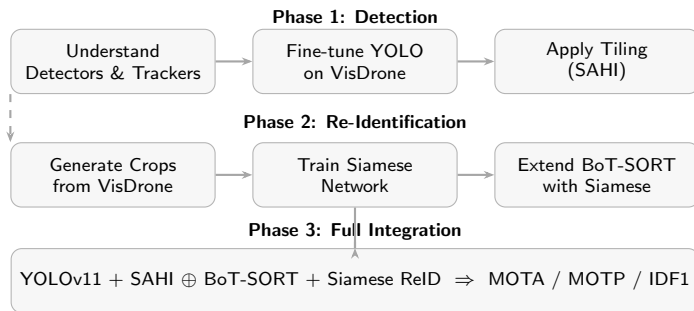
- **UAV Constraints:** High altitude and scale variance render standard CNN receptive fields inadequate.
- **Architectural Evolution:** Feature Pyramid Networks (FPN) and YOLO11 leverage multi-scale semantic merging.
- **Tiling (SAHI):** Image slicing prevents downsampling loss, preserving high-resolution features for small targets.

Temporal Tracking & Re-ID

- **Association:** Occlusions and crowds necessitate Siamese-based Re-Identification (Re-ID).
- **Metric Learning:** Contrastive and Focal-BCE losses optimize distance-based embedding discrimination.
- **Optimization:** Optuna (TPE) replaces grid search for efficient Bayesian hyperparameter tuning.

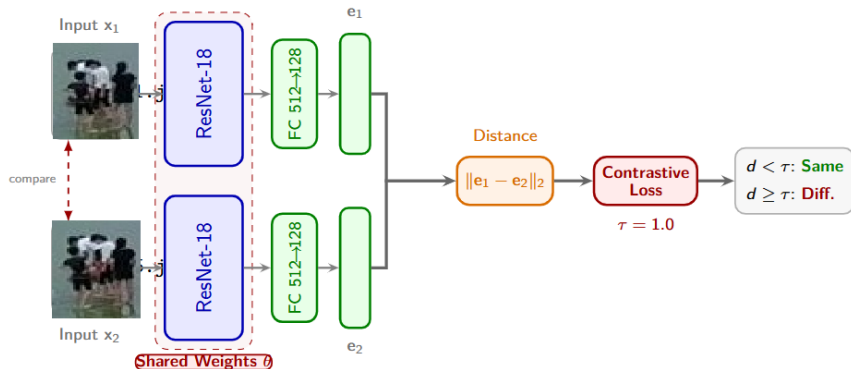
The Proposed Pipeline

SAHI + Fine-tuned YOLO11 + Optimized Siamese Re-ID + BoT-SORT





Metric	YOLOv8n (Siamese NW)	YOLO11 (FT)	YOLO11 + SAHI	YOLO11 + SAHI + BotSort (380, 0.3)
MOTA	0.129	0.1673	0.1728	0.0740
IDF1	0.224	0.4902	0.4879	0.4382
HOTA	0.260	0.2124	–	–
DetA	0.142	0.3014	–	–
AssA	0.504	0.1557	–	–
MOTP	0.222	0.2651	0.2601	0.2578
Precision	0.8168	–	0.6112	0.5491
Recall	0.1298	–	0.4907	0.4456
FPS	12.69	0.5	0.5	–





Key Findings:

- **Best Model (Run 12):** ResNet18 ($d = 64$, $m = 1.2$) achieved **0.989 F1** with high discriminative separation ($\approx 20.6\times$ distance ratio).
- **End-to-End Training:** Freezing backbones (Run 4) dropped F1 to 0.829; fine-tuning is vital for UAV imagery.
- **Efficiency:** ResNet18 matched ResNet34 accuracy but trained **$2.4\times$ faster**, making it the optimal real-time choice.

Table: Top-5 Grid Sweep Runs

Run	Backbone	d	m	F1
12	ResNet18	64	1.2	0.989
10	ResNet34	64	0.8	0.988
1	MobileNetV3-S	256	0.8	0.964
5	ResNet18	256	1.0	0.906
4	ResNet18*	64	1.2	0.829

* Frozen backbone. All others unfrozen with L2 norm.

(a) Baseline (MobileNetV3-S, $d = 256$, $\text{lr} = 10^{-4}$, $m = 1.0$)

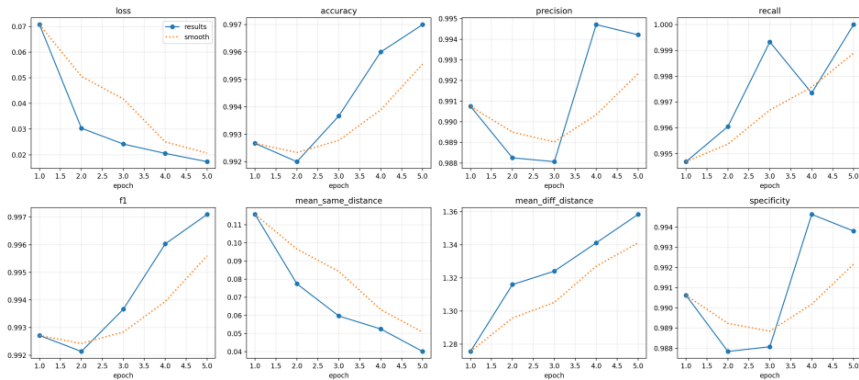
Epoch	Loss	Acc	Prec	Rec	F1
1	2.660	0.660	0.600	0.970	0.740
2	0.210	0.670	0.610	0.990	0.750
3	0.130	0.710	0.630	1.000	0.770
4	0.090	0.790	0.700	1.000	0.820
5	0.080	0.770	0.680	1.000	0.810

(b) Optimised (ResNet18, $d = 64$, $\text{lr} = 3 \times 10^{-4}$, $m = 1.2$, L2 norm)

Epoch	Loss	Acc	Prec	Rec	F1
1	0.071	0.993	0.991	0.995	0.993
2	0.030	0.992	0.988	0.996	0.992
3	0.024	0.994	0.988	0.999	0.994
4	0.021	0.996	0.995	0.997	0.996
5	0.017	0.997	0.994	1.000	0.997

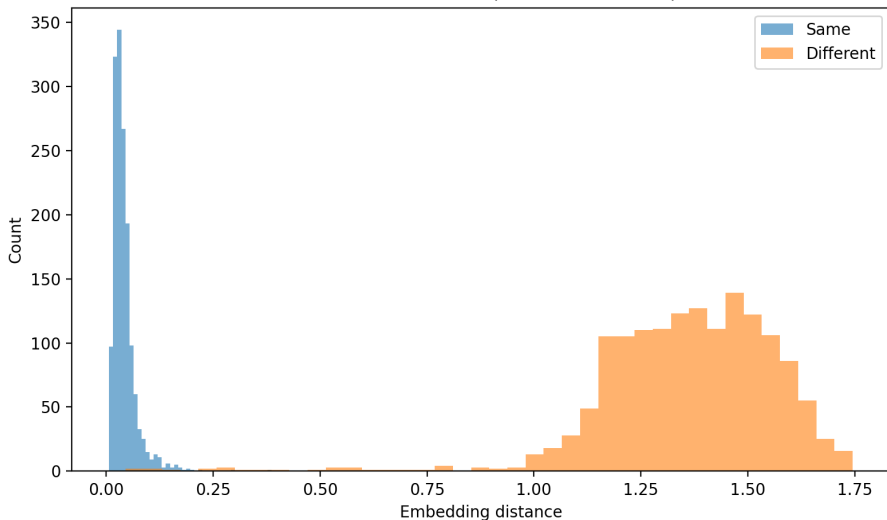
Training dynamics

Optimised Siamese network



Same vs Different Object Distance

Distance Distribution (Same vs Different)



Hyperparameter & Threshold Analysis

Distance Threshold (τ) Sweep

τ	Acc	Prec	Recall	F1
0.5	0.784	0.838	0.693	0.759
0.8	0.776	0.740	0.851	0.791
1.0	0.762	0.690	0.916	0.802
1.2	0.706	0.632	0.959	0.762
1.5	0.517	0.500	0.995	0.666

- **High Precision:** $\tau = 0.5$ minimizes false ID matches.
- **Optimal F1:** $\tau = 1.0$ provides the best overall balance.
- **Saturation:** $\tau > 1.5$ leads to random-guess precision.

Optuna Bayesian Optimization

Trial	d	w_{neg}	L2	F1
7	64	2.03	No	0.844
2	64	2.31	Yes	0.843
8	64	2.60	Yes	0.835
6	128	2.26	Yes	0.792
0	128	2.81	Yes	0.786

- **Best Settings:** Focal-BCE with $w_{\text{neg}} \approx 2.0$ and $d = 64$.
- **Backbone:** End-to-end fine-tuning consistently outperformed frozen backbones.
- **L2 Norm:** Top trial succeeded without L2 normalization.

Alternative Loss Comparison

- **Objective:** Compare standard Focal-BCE (*Baseline*) against a precision-weighted variant (*Enhanced*) to minimize identity switches in tracking.

Table: Focal-BCE Variant Comparison (5 Epochs)

Variant	w_{neg}	Acc	Prec	Recall	F1	FP
Baseline	1.0	0.787	0.709	0.964	0.817	586
Enhanced FP-Red	2.0	0.808	0.753	0.924	0.830	460

Enhanced Advantage

- **False Positive Reduction:** Dropped from 586 to 460 (−21.5%).
- **Improved Precision:** Increased to 0.753, crucial for reducing spurious track initializations.

The Trade-off

- **False Negatives:** Increased from 53 to 115 as the model becomes more "conservative."
- **Verdict:** Enhanced variant is preferred for BoT-SORT integration to protect **MOTA** scores.

- **Comparison:** Baseline YOLO11 vs. SAHI-tiled inference across 7 validation sequences.

Table: Pipeline Comparison (VisDrone2019-MOT Val)

Metric	Prelim. (CPU)	YOLO+Siamese	SAHI+Siamese
MOTA ↑	0.129	0.326	0.061
IDF1 ↑	0.224	0.504	0.453
HOTA@0.5 ↑	0.260	0.495	0.429
AssA@0.5 ↑	0.504	0.629	0.532
FP ↓	35,629	11,374	52,478
ID Switches ↓	657	578	1,405
FPS ↑	12.69	5.84	1.27

Key Observation

While SAHI is intended for small objects, it introduced a massive surge in **False Positives (FP)** and **ID Switches**, leading to a significant MOTA degradation compared to the standard YOLO11-fine-tuned pipeline.

BoT-SORT Hyperparameter Tuning

Goal

- Systematically improve tracking performance using a 4-phase tuning strategy

Evaluation Strategy

- All trials logged for reproducibility
- Composite score:
 - $0.45 \times \text{IDF1}$
 - $0.35 \times \text{HOTA}$
 - $0.20 \times \text{MOTA}$
- Focus: identity consistency

Phase 1: Association Tuning (10 trials)

- Parameters: appearance_weight, match_cost_thresh, IoU thresholds
- Best configuration:
 - appearance_weight = 0.65
 - match_cost_thresh = 0.72
- Key finding: Optimal balance between appearance and spatial matching achieved at these values

Phase 2: Track Lifecycle (24 trials)

- Parameters: min_hits, max_age, new_track_thresh
- Key finding: min_hits = 1 is optimal since UAV objects appear briefly and require immediate track confirmation

Phase 3: Kalman Filter (Pending)

- Tune motion model and noise parameters

Phase 4: Reassociation (Pending)

- Recover lost tracks
- Reduce fragmentation

YOLO vs. SAHI Comparison

- **CMC Integration:** Sparse optical flow CMC implemented to mitigate UAV ego-motion and inflate IoU accuracy.
- **Optimization Progress:**
 - **Phases 1–2 (Complete):** Calibration of gates/lifecycle; min_hits=1 identified as critical.
 - **Phases 3–4 (Pending):** Kalman dynamics and local reassociation remain incomplete.

Per-Sequence MOTA / IDF1 / HOTA Breakdown

Sequence ID	YOLO11 + Siamese (Ours)			SAHI + Siamese		
	MOTA ↑	IDF1 ↑	HOTA ↑	MOTA	IDF1	HOTA
uav0000086	0.361	0.512	0.469	0.295	0.499	0.446
uav0000117	0.280	0.473	0.474	-0.532	0.348	0.360
uav0000137	0.367	0.539	0.515	0.165	0.494	0.421
uav0000182	0.324	0.526	0.485	0.158	0.507	0.447
uav0000268	0.150	0.270	0.318	0.018	0.267	0.313
uav0000305	0.512	0.705	0.682	-0.056	0.565	0.560
uav0000339	0.348	0.509	0.522	0.221	0.469	0.454
Overall	0.326	0.504	0.495	0.061	0.453	0.429



Component	Refinement	Expected gain
SAHI tiling (detection)	Per-sequence tiling policy: sweep slice size $\{384, 512, 640\} \times$ overlap $\{0.2, 0.3, 0.4\}$; class-aware NMM threshold (tighter for cars, looser for pedestrians); edge-padded slices to avoid boundary truncation.	Recall $\uparrow$ on 20–40 px objects; fewer boundary fragments
Siamese ReID + NMS (appearance)	Embedding-aware NMS: before tracker input, suppress duplicate boxes whose cosine-sim $> \tau_e$ and IoU $> \tau_i$ (survives a bbox-only NMS collision). Temperature-scale embeddings; L2-normalise before cache write.	Fewer duplicate tracks; cleaner ID assignment
BotSort hyper-params (association)	Grid-search on val: track_high $\in \{0.5, 0.6\}$, new_track $\in \{0.6, 0.7\}$, match $\in \{0.7, 0.8\}$, appearance $\in \{0.2, 0.25\}$, proximity $\in \{0.4, 0.5\}$, buffer $\in \{30, 45, 60\}$; CMC warp $\rightarrow$ affine <i>before</i> predict (not after).	MOTA $\uparrow$, IDs $\downarrow$, IDF1 $\uparrow$

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- Wang, J. et al. (2021). *Tiny object detection in aerial images*. ICPR.
- Liu, M. et al. (2020). *UAV-YOLO: Small object detection from UAV perspective*. Sensors.
- Feng, F. et al. (2024). *Improved YOLOv8 for small object detection in aerial imagery*. Journal of King Saud University.